THE MAXIMAL OPEN WHERE THE STABILIZER HAS A TAU-LOCAL PROPERTY IS R-INVARIANT

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Allowed standard prerequisites (editorial).

- R-invariant opens of a groupoid: W with t^{-1}(W) = s^{-1}(W) (Groupoids chapter, Definition groupoids-definition-invariant-open, canonical tag 03BC). cited at statement lines 687-690. States the target notion: W is R-invariant exactly when its inverse images under s and t agree.
- Largest open of the base on which a tau-locally checked morphism property holds, with its pullback characterization over base changes (Descent chapter, Lemma descent-lemma-largest-open-of-the-base, canonical tag 06QP). cited at proof lines 694-695 and 702-704. Gives the maximal open W for G_W to W and, applied to the isomorphic base changes of G to U along t and along s, the equality t^{-1}(W) = s^{-1}(W).

Source reference. Original reference 03BC: sources/stacks-chow/tags.txt, source lines 4343–4343. Cited at statement lines 687-690: W is R-invariant, meaning t^{-1}(W) = s^{-1}(W).

Source reference. Original lemma 06QP: sources/stacks-chow/tags.txt, source lines 8613–8613. Cited at proof lines 694-695 and 702-704: existence and maximality of W, and the pullback characterization giving t^{-1}(W) = s^{-1}(W).

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid over S. Let $G \to U$ be its stabilizer group scheme. Let $\tau \in \{fppf, étale, smooth, syntomic\}$. Let $\mathcal{P}$ be a property of morphisms which is τ-local on the target. Assume $\{s : R \to U\}$ and $\{t : R \to U\}$ are coverings for the τ-topology. Let $W \subset U$ be the maximal open subscheme such that $G_W \to W$ has property $\mathcal{P}$. Then W is R-invariant (see Groupoids, Definition 03BC).*

Proof. The existence and properties of the open $W \subset U$ are described in Descent, Lemma 06QP. The morphism

$$G \times_{U,t} R \longrightarrow R \times_{s,U} G, \quad (g, r) \longmapsto (r, r^{-1} \circ g \circ r)$$

is an isomorphism over R (where $\circ$ denotes composition in the groupoid). Hence $s^{-1}(W) = t^{-1}(W)$ by the properties of W proved in the aforementioned Descent, Lemma 06QP. □

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